# Fizika

9 sinf

## MAVZU: Masalalar yechish

# Modda miqdori

Moddadagi molekulalar (atomlar) sonining 12 g ugleroddagi atomlari soniga nisbati.

$$\nu = \frac{m}{M} = \frac{N}{N_A}$$

# Avogadro qonuni

Har qanday moddaning 1 molida $6 \cdot 10^{23}$ ta molekula (atom) mavjud.

Har qanday gazning  1 moli normal sharoitda 22,4 l hajmni egallaydi.

# Molyar massa

Miqdori 1 mol bo‘lgan moddaning massasi

$$M = m_0 \cdot N_A$$

Mendeleyev davriy sistemasiga asosan aniqlanadi.

Molyar konsentratsiya.

Hajm birligidagi molekulalar soni.

$$n = \frac{N}{V}$$

$$m = m_0 \cdot N$$



$$m = \frac{M}{N_A} \cdot N$$



$$N = \frac{m}{M} \cdot N_A$$

# 1-namunaviy masala

Uzunligi 15 sm va ko‘ndalang kesim yuzi $4\,mm^2$ bo‘lgan grafit qalamchasida qancha uglerod atomi borligini aniqlang. Grafitning zichligi $1{,}6\,\frac{g}{sm^3}$. Bir dona uglerod atomining massasi $2 \cdot 10^{-26} kg$ ga teng.

## Berilgan:

$$l = 15\ sm = 15 \cdot 10^{-2}\ m$$

$$S = 4\ mm^2 = 4 \cdot 10^{-6} m^2$$

$$\rho = 1{,}6\ \frac{g}{sm^3} = 1{,}6 \cdot \frac{10^3 kg}{m^3}$$

$$m_0 = 2 \cdot 10^{-26} kg$$

**Topish kerak:** $N - ?$

## Formula:

$$N = \frac{m}{m_0}, \quad m = \rho \cdot V$$

$$V = S \cdot l$$

$$N = \frac{m}{m_0} = \frac{\rho \cdot V}{m_0} = \frac{\rho \cdot S \cdot l}{m_0}$$

## Yechish:

$$N=\frac{1{,}6\cdot\frac{10^3\,kg}{m^3}\cdot 4\cdot 10^{-6}m^2\cdot 15\cdot 10^{-2}\,m}{2\cdot 10^{-26}kg}=4{,}8\cdot 10^{22}ta$$

**Javob: $N= 4{,}8\cdot 10^{22}ta$**

# 3-mashq 3-masala

Idishga solingan gazning massasi $5{,}5\,kg$ ga teng. Idishda $7{,}5 \cdot 10^{25}$ ta gaz molekulasi mavjud bo‘lsa, bu gaz turini aniqlang.

## Berilgan:

$m = 5{,}5\ kg$

$N = 7{,}5 \cdot 10^{25} ta$

$N_A = 6 \cdot 10^{23} mol^{-1}$

## Topish kerak: $M -?$

## Formula:

$$N = \frac{m}{M} \cdot N_A$$

$$M = \frac{m \cdot N_A}{N}$$

## Yechish:

$$M = \frac{5{,}5\ kg \cdot 6 \cdot 10^{23} mol^{-1}}{7{,}5 \cdot 10^{25}} = 44 \frac{10^{-3} kg}{mol} == 44 \frac{g}{mol}$$

**Javob: $CO_2$(karbonat angidrid)**

# 3-mashq 8-masala

Modda miqdori 34 mol simob qancha hajmni egallaydi? Simobning zichligi $13{,}6\frac{g}{sm^3}$, molyar massasi 200 $\frac{g}{mol}$ deb oling.

## Berilgan:

$$\nu = 34\ mol$$

$$\rho = 13{,}6\,\frac{g}{sm^3} = 13600\,\frac{kg}{m^3}$$

$$M_{(Hg)} = 200\,\frac{g}{mol} = 0{,}2\,\frac{kg}{mol}$$

## Formula:

$$m = \rho \cdot V \Rightarrow V = \frac{m}{\rho}$$

$$\nu = \frac{m}{M} \Rightarrow m = \nu \cdot M$$

$$V = \frac{\nu \cdot M}{\rho}$$

**Topish kerak:** $V - ?$

**Yechish:** $V = \frac{34\ mol \cdot 0{,}2\ kg/mol}{13600\ kg/m^3} = 0{,}0005\ m^3 = 0{,}5\ l$

**Javob:** $V = 0{,}5\ l$

# Mustaqil bajarish uchun topshiriqlar

3-mashq (17-bet) 1,2,5,6-masalalarni yechish.